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***Abstract*—The growing use of wearable devices has increased interest in egocentric AI, while the software infrastructure linking hardware data with high-level models remains fragmented. This paper presents Lucia, a wearable platform consisting of a compact pin device and a dedicated Software Development Kit (SDK) for egocentric research. The Lucia SDK provides a unified interface for data acquisition via Android Debug Bridge (ADB) and wireless communication, and supports real-time RTSP streaming for low-latency applications. We validate the platform through two case studies: (A) integration with the ST-Think model for spatio-temporal reasoning, enabling 3D path planning and natural-language question answering on egocentric video; and (B) a binocular stereo pipeline combining Semi-Global Block Matching with Depth Anything V2 and Large Language Model(LLM) for metric depth estimation and object size measurement. Results show that Lucia simplifies the deployment of advanced vision models on wearable hardware. The SDK is publicly released to support reproducible research in wearable computing.**

***Index Terms*—Egocentric AI, SDK, ST-Think model, Stereo**

I. Introduction

The emergence of wearable computing devices has fundamentally transformed how humans interact with artificial intelligence systems in daily life. Smart glasses and wearable cameras, in particular, have evolved from niche research prototypes to mainstream consumer products capable of capturing rich egocentric data streams [1], [2]. These devices offer a unique first-person perspective that enables AI systems to perceive the world as the user experiences it, opening new possibilities for context-aware assistance, navigation, and environmental understanding [3], [4].

Recent advances in miniaturized hardware have enabled the development of compact wearable devices equipped with high-resolution cameras, inertial measurement units, and wireless communication capabilities [5]–[7]. Smart glasses for visually impaired individuals exemplify this trend, integrating computer vision, deep learning, and real-time audio feedback to enhance mobility and independence [8]–[10]. Concurrently, research on augmented reality systems has demonstrated the potential of head-mounted displays for applications ranging from industrial inspection to medical training [11]–[13]. The integration of large language models with wearable cameras has further expanded the scope of intelligent assistants, enabling natural language interaction with visual scene understanding [14], [15].

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Despite these hardware advances, the software infrastructure supporting wearable device development remains fragmented. Developers working with smart glasses and wearable cameras face significant challenges in device connectivity, cross-platform compatibility, and real-time data streaming [16], [17]. The Android Open Source Platform (AOSP) has emerged as a common foundation for many wearable devices, yet unified SDK solutions that abstract hardware complexities while enabling advanced AI applications are lacking [16]. Furthermore, the transition from proprietary development environments to open-source platforms is essential for fostering academic research and reproducible experimentation [18].

Egocentric video analysis has emerged as a vibrant research area, driven by large-scale datasets and sophisticated multi-modal models [1], [4], [19]. Applications span from hand-object interaction detection for rehabilitation assessment [20]–[22] to activity recognition for smart environments [23]–[25]. Navigation assistance represents a particularly compelling use case, where AI systems must combine spatial reasoning, temporal memory, and natural language generation to guide users through complex environments [15], [26], [27]. Synthetic data generation has also proven valuable for training egocentric perception models [3], while edge computing techniques enable efficient on-device inference [28]–[30].

Beyond navigation, stereo vision and depth estimation from wearable devices have attracted considerable attention for applications in augmented reality, robotics, and assistive technology [31], [32]. The fusion of classical geometric methods with modern deep learning approaches has yielded hybrid pipelines that combine the metric accuracy of stereo matching with the dense predictions of monocular depth networks [33], [34]. Object detection and size estimation through wearable cameras enable intelligent scene understanding that supports both retrospective analysis and real-time interaction [35]–[37].

In this paper, we present Lucia, a compact wearable pin device accompanied by a comprehensive Software Development Kit (SDK) designed to facilitate both research and application development. The Lucia Pin captures high-resolution egocentric video and connects to host systems through multiple modalities: wired USB connections via Android Debug Bridge (ADB) for development workflows, and wireless Bluetooth and Wi-Fi bridges for untethered operation. Our SDK provides a unified programming interface that abstracts connectivity details, enabling developers to focus on higher-level AI applications rather than low-level device management.

We demonstrate the capabilities of the Lucia platform through two case studies that showcase representative AI-enabled applications. First, we present a navigation task that combines egocentric video captured by Lucia with a pretrained spatio-temporal reasoning model (ST-Think) for 3D path-planning and question answering. This application addresses retrospective queries about past movements and prospective queries about navigation routes, illustrating how wearable egocentric data can support environment-level reasoning. Second, we develop a stereo reconstruction system using a custom dual-Lucia rig that enables binocular depth estimation and object size measurement. By fusing sparse geometric depth from Semi-Global Block Matching with dense relative depth from Depth Anything V2, and leveraging LLM-guided object detection through YOLO-World and SAM, we demonstrate an end-to-end pipeline for metric 3D perception.

The main contributions of this paper are summarized as follows:

- We introduce the Lucia SDK, a comprehensive software framework that provides unified device connectivity through ADB and wireless protocols, file management through an intuitive GUI, and real-time video streaming via RTSP.
- We present a plug-and-play integration with ST-Think for navigation-oriented question answering over egocentric recordings, demonstrating that minimal engineering effort is required to enable environment-level reasoning.
- We develop a stereo Lucia platform with calibration, depth fusion, and LLM-guided object classification pipelines that achieve accurate metric depth estimation and physical size measurement.
- We make our SDK publicly available as an open-source platform to support reproducible research and accelerate the development of AI applications on wearable egocentric devices.

The remainder of this paper is organized as follows. Section **??** describes the connectivity architecture of the Lucia Pin, including ADB connection, wireless bridging, and RTSP streaming capabilities. Section III presents two case studies: the navigation task with path-planning question answering and the reconstruction task with stereo depth estimation and object measurement. Section IV concludes the paper with a discussion of limitations and future directions.

II. CONNECTIVITY OF LUCI PIN

A. ADB Connection

The Lucia Pin can be connected to a personal computer via Android Debug Bridge (ADB) connection over USB. When connected, the system provides information on whether the connection is successful, and allows developers to issue commands via ADB Shell. Through ADB Shell, developers can view the device status, access files on the pin, transfer files between the pin and PC, and configure networks to connect the pin to nearby hotspot connections. As such, developers and users no longer need to set up custom USB drivers. ADB Shell also allows developers to build additional connectivity features. Key features supported by the current ADB Connection are summarized as follows:

a) Device Discovery and Basic Information Retrieval: When the device is connected via USB, the SDK allows ADB connection to be established. The ADB connection automatically initiates the device discovery process where all USB-connected ADB devices are enumerated. Active devices are detected and the first Lucia Pin is selected. The system then stores the device ID in the LUCI instance and will use it throughout the active session.

Basic information of the device can also be retrieved via the SDK. Information on how much storage space is used and how much remains allow users to assess the remaining recording capacity, and information on the Linux OS metadata of the pin can also be accessed. If the device was previously connected to a hotspot, a cached IP address can also be accessed if it is available. These features to access basic device information are implemented in the DeviceAPI layer by calling ADB Shell commands.

b) File Browser and File Transmission: The SDK provides a script to browse and transfer files via a Graphical User Interface (GUI). All features such as detecting and listing directories and navigating within the file system are implemented by executing ADB Shell commands on the pin device. File transmission is achieved with ADB push and pull commands.

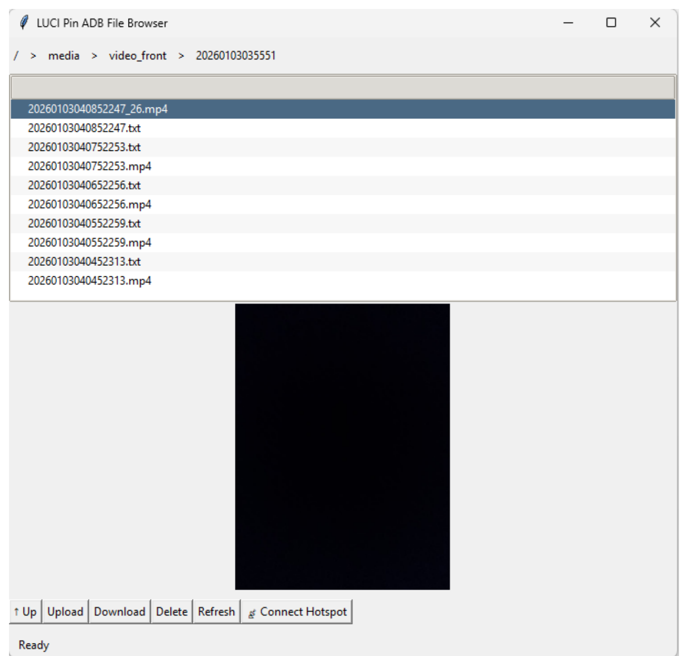


Fig. 1. Graphical User Interface of the Lucia Pin file browser and transmission system.

These operations are packaged with the FileBrowserGUI class. This class provides an interactive interface similar to the traditional file explorer interface, as shown in Figure 1. The GUI allows users to navigate within the file system in the Lucia pin by clicking on breadcrumbs in the top panel, subfolders in the main window, and the up button in the bottom panel. Users can also upload files from the PC to the pin and download selected files from the pin to a local directory in the PC by clicking buttons in the bottom panel. The bottom panel also allows users to delete selected files, refresh folder contents.

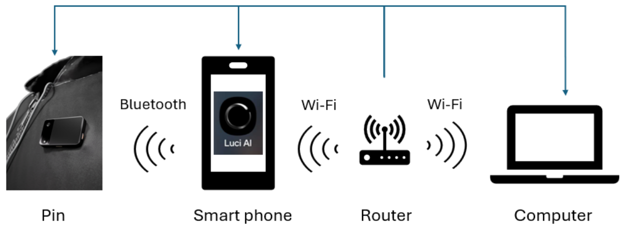


Fig. 2. The Lucia Pin connects to the smartphone via Bluetooth, which then bridges the device onto the Wi-Fi network, enabling IP-based access from a computer on the same router.

c) Wireless Network Configuration: The GUI also allows users to connect to a hotspot hosted by another device directly by clicking a button in the bottom panel in Figure 1 and providing the hotspot name and password. Developers can also connect the pin to a hotspot by providing the hotspot name and password via python commands implemented in the LUCI class. After the hotspot configuration script is executed successfully on the pin device, the SDK attempts to detect the newly assigned IP address. This detected IP address is cached within the LUCI instance which facilitates subsequent actions such as viewing the pin video livestream via RTSP on the hotspot host machine.

B. Wireless Connection

Lucia enables seamless interaction between users and its AI capabilities through multiple methods: hand gestures, voice input akin to Siri, direct input via the hub's interface, and remote control using a mobile phone. These diverse interaction options allow users to intuitively control device functions and access AI features on both the hub and connected mobile devices, enhancing the overall user experience.

The hub is equipped with Bluetooth connectivity, allowing it to connect seamlessly with other wearable smart devices such as smartwatches, AR glasses, and wireless earphones. This feature supports a versatile range of inter-device connections, enabling the hub to control other smart devices and, with appropriate permissions, allowing those devices to control the hub as well.

The Lucia Pin also connects to the smartphone via the Luci AI application using a Bluetooth link as shown in Fig. 2. The smartphone then acts as a network bridge, placing the Pin onto the local Wi-Fi network and allowing it to obtain an IP address from the router. When a computer joins the same network, the Pin becomes directly reachable through its assigned IP. In this configuration, the smartphone can be used to view the pin camera, while the computer can access the RTSP stream of the device to enable real-time capture, monitoring and recording of videos. These features are based on FFmpeg and RTSP and are all supported by the SDK.

When the pin is connected to the hotspot of the host machine, the SDK allows users to view the livestream of the pin camera. Upon providing a keyboard input, the current frame can be saved in a local directory. Real-time livestream can also be achieved by accessing the RTSP stream by providing the pin IP address in an external video media player software. Similarly, based on the RTSP stream, the SDK also allows the livestream video to be saved in a local directory with the option to specify the segment time.

III. Case Study: Exploring Lucia's Real-world Applications

A. Navigation Task: 3D Path-Planning Question Answering

1) Task Definition and Motivation: Lucia is a compact wearable device that can record rich egocentric data and naturally captures fine-grained details of everyday life. Given the data captured by Lucia, we construct a base task framework that understands a basic 3D environment and answers queries of the form "What path did I take from A to B?" or "How do I get from A to B?" by producing a logically consistent explanation in natural language.

In real-life scenarios, AI assistants typically arise from the need to remember and recall detailed information that would otherwise be forgotten, which is precisely the capability we aim to explore. To resolve such path-related queries, two ingredients are essential: (i) a basic understanding of the environment, and (ii) a temporal memory of the user's movement captured from the egocentric perspective. To answer the questions in natural language, it is crucial to combine localization with temporal retrieval. In our experiment, we connect Lucia to a plug-and-play pretrained model for question answering over the reconstructed trajectory and scene memory. This setup demonstrates that, with minimal additional engineering, Lucia can already support environment-level reasoning about both past movements and plausible future routes.

2) System Pipeline: In this project, Lucia acts as the interaction interface, and we employ a pretrained model for basic environment understanding and logical retrieval. We choose ST-Think, a video-based multimodal model specifically designed for spatio-temporal reasoning over egocentric "4D world" videos. The pipeline is structured as follows.

3) Data Capture with Lucia: Our pin device is worn on the chest, making it well-suited for egocentric video capture and convenient to use in both indoor and outdoor environments. After recording, the full sequence is available in standard video format, which can be easily loaded for subsequent processing. This enables straightforward operations such as video segmentation, frame extraction, and feeding selected segments into reconstruction or higher-level understanding modules.

4) Prompt Construction: For this task, we use ST-Think as the core model for all downstream reasoning. ST-Think is a 4D world understanding model that takes video as input and answers questions specified in a textual prompt. In our setup, the model processes the selected Lucia video segment and responds to the question embedded in the prompt. Since ST-Think is formulated as a question-answering model that selects or generates the best answer from candidate options, it is important to design prompt settings that are well aligned with the evaluation in this experiment.

Concretely, we construct a short textual prompt that combines the user's question with a minimal list of anchors (e.g., named locations or landmarks) that can be used as reference points in the answer choices. We then format the input in ST-Think's QA style, which consists of the video segment paired with the question, and optionally request step-by-step reasoning to encourage explicit path descriptions.

5) Experiment Setup: We evaluate our Lucia+ST-Think pipeline on egocentric recordings collected in a small set of everyday indoor spaces, including an office corridor, a lab, and a studio-style apartment. For each environment, a wearer follows several predefined routes that connect different functional areas (e.g., entrance, desk, kitchen), yielding multiple start–end pairs with varying route length and complexity. The resulting Lucia videos form the input to ST-Think without any task-specific fine-tuning.

From these recordings we construct two types of questions: (i) *retrospective* queries that ask the model to describe the path actually taken between two locations, and (ii) *prospective* queries that ask how to go from one location to another within the same environment. Following the categories used in ST-Think's original benchmark, we focus on route description and direction-change understanding, and phrase questions with mild lexical variation to test robustness.

Consistent with the benchmark design, direction-change understanding is evaluated using exact-match accuracy at key junctions, while route descriptions are assessed using an open-ended evaluation protocol that emphasizes logical consistency and alignment with the ground-truth path.

We compare three model variants: a text-only baseline that receives only a symbolic room list without video, a generic video LMM baseline, and ST-Think as our main 4D video reasoning model. Performance is reported in a result table using path correctness (sequence agreement with the ground-truth route), direction-change accuracy (left/right/straight at key junctions), and a human-rated "followability" score that assesses whether the generated instructions are clear enough to be followed in practice. For open-ended responses, we adopt an LLM-judge workflow adapted from ST-Think, complemented by human ratings to assess instruction followability in realistic navigation scenarios.

6) Quantitative Results: Table **??** reports the quantitative performance of our Lucia+ST-Think pipeline compared to the text-only baseline and a generic video LMM. Overall, ST-Think achieves the highest path correctness on both retrospective and prospective queries, with a clear margin over the text-only model. This indicates that access to the raw egocentric video, combined with 4D spatio-temporal reasoning, is crucial for reconstructing and explaining the user's route.

On the direction-change metric, ST-Think also shows consistent gains, producing more accurate left/right/straight decisions at junctions than both baselines. Human raters further judge its step-by-step instructions to be more "followable", assigning higher clarity scores and noting that the presence of visual landmarks (e.g., doors, furniture, or distinctive corners) makes the guidance easier to execute in practice.

To complement the end-to-end path QA scores in Table **??**, we further report a category-level breakdown that isolates the underlying perception and reasoning components. Table **??** summarizes accuracy, error rate, and confidence across two representative settings: a visual-perception "Beach" probe (color and coarse location) and a navigation reasoning probe (spatial relations, temporal ordering, object reference, and long-horizon navigation). This breakdown helps explain where errors originate (e.g., appearance vs. reasoning) and highlights the relative difficulty of each category.

As shown in Table **??**, perception-style categories achieve higher accuracy and confidence (e.g., Beach color/location at 88.5–90.5% accuracy with $\sim$89–96% confidence), indicating that the model is generally reliable on direct visual attributes under favorable conditions. In contrast, navigation-oriented categories are markedly more challenging: spatial and temporal reasoning drop to 75.5% and 81.5% accuracy, while long-horizon navigation shows the highest error rate (34.0%), consistent with accumulated uncertainty over multi-step planning. The dominant error types further suggest actionable improvement directions—lighting sensitivity for appearance cues, viewpoint ambiguity for coarse localization, and long-horizon reasoning failures for multi-turn navigation—aligning with the qualitative feedback from raters regarding landmark use and instruction followability.

Taken together, these results suggest that ST-Think is an effective core engine for path question answering on Lucia recordings: it not only recovers the underlying route structure more reliably than simpler baselines, but also expresses the path in language that users can readily follow.

7) Discussion and Limitations: This case study shows that combining Lucia's always-on egocentric capture with ST-Think's 4D video reasoning provides a practical pipeline for path question answering. Without task-specific fine-tuning, the model can already turn raw wearable recordings into step-by-step route descriptions that align reasonably well with the user's actual movement and are understandable to human readers.

However, our current setup has several limitations. First, both video preprocessing and ST-Think inference are run offline, so the system does not yet support real-time guidance. Second, the evaluation is limited to a small number of relatively simple indoor environments, and we observe occasional errors in visually similar corridors or near identical doors. Third, we rely only on the egocentric video stream, without explicitly integrating richer 3D reconstructions or environment maps that Lucia can provide elsewhere in our pipeline.

Future work will focus on tightening this integration: enriching ST-Think's prompts with structured 3D and semantic summaries from Lucia's memory, scaling to larger and more diverse spaces, and progressively moving the computation towards near real-time guidance on the user's device.

B. Two-View Vision Experiment: Stereo Depth Estimation and 3D Reconstruction.

The Lucia pin is a monocular visual device with high resolution and flexible video formats, as listed in Table. **??**. In practical use, however, a single camera cannot provide depth information on its own. To understand whether Lucia can support 3D perception in near-range tasks, we designed a two-view vision experiment based on a stereo configuration of the

TABLE I
EXPERIMENT SETUP AND EVALUATION PROTOCOL

Category	Setting	Value
Input modality	Egocentric video	Lucia RGB video only
Environments	Indoor scenes	Office corridor, lab, studio apartment
Route design	Start–end pairs	Multiple routes per scene with varying length
Task types	Query types	Retrospective and prospective path queries
Task focus	Reasoning aspects	Route description, direction-change understanding
Question format	Prompting	Zero-shot, natural language questions
Lexical variation	Robustness	Mild paraphrasing across queries
Models evaluated	Compared systems	Text-only baseline; generic video LMM; ST-Think
Prompting regime	Inference setting	Zero-shot, default cue per model
Metrics (MCQ-style)	Direction changes	Accuracy (exact match)
Metrics (open-ended)	Path description	LLM-judge scoring with route evaluation prompt
Human evaluation	Followability	Clarity and executability of instructions
Training	Fine-tuning	None (test-only evaluation)

TABLE II
UNIFIED CATEGORY-WISE PERFORMANCE SUMMARY (BEACH ANALYSIS + NAVIGATION BENCHMARK)

Cat.	Tot.	Corr.	Incorr.	Acc. (%)	Err. (%)	Conf. (%)	Time/Q (s)	Err. Type	Diff.
Color (Beach)	200	181	19	90.5	9.5	96.3	398	Lighting bias	Easy
Location (Beach)	200	177	23	88.5	11.5	89.4	395	Viewpoint ambiguity	Medium
Spatial (Nav)	200	151	49	75.5	24.5	79.2	529	Spatial relation ambiguity	Hard
Temporal (Nav)	200	163	37	81.5	18.5	83.6	496	Temporal ordering error	Hard
Object (Nav)	200	169	31	84.5	15.5	88.1	570	Object confusion	Medium
Navigation (Nav)	200	132	68	66.0	34.0	78.4	816	Long-horizon reasoning failure	Hard

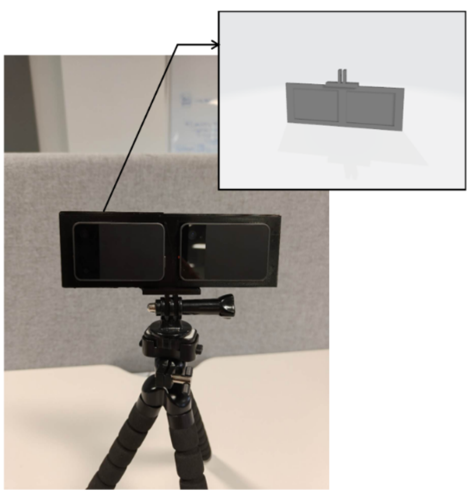


Fig. 3. CAD rendering (top-right) and physical assembly of the Lucia Pin stereo platform.

device. This setup allows us to study stereo depth estimation and 3D reconstruction using lightweight monocular sensors.

1) Stereo Lucia Capture: By mounting two Lucia pins onto a rigid fixture (Fig. **??**), we created a simple stereo platform of the device as shown in Fig. **??**. This setup allows us to examine how well Lucia performs in estimating depth and surface geometry in scenarios that require accurate distance measurements. The custom stereo platform serves as a 3D printed lightweight testbed for evaluating near-range depth estimation and its potential use in future visual perception tasks.

As mentioned in Section. **??**, the two Lucia pins were connected to the same Wi-Fi router and configured with the IP addresses. Stereo image pairs were obtained by retrieving frames directly from their RTSP streams. This networked setup enabled flexible access to live camera views from any device on the network, while providing a practical way to collect paired multi-view observations for depth reconstruction. Importantly, capturing frames from the RTSP streams also supported frame-level synchronization, which was essential for recording consistent stereo pairs and for analyzing calibration results and depth accuracy.

2) Monocular Calibration Parameters: Accurate camera intrinsics are required for the stereo calibration process. Lucia is equipped with a 12-megapixel monocular visual sensor. The intrinsic parameters of this camera are calibrated using OpenCV's [38] implementation of Zhang's method [39]. As illustrated in Fig. **??**, a chessboard pattern with 8×5 inner corners is recorded across 48 images covering diverse orientations and spatial positions. The resolution of the camera is 960×1280 pixels (width × height).

The detected corner features are shown in Fig. **??** by using Opencv. The camera is modeled using the standard pinhole projection formulation, with calibration estimating a 4-parameter intrinsic matrix and a 5-coefficient Brown–Conrady distortion model [40]. Table. III details the calibrated parameters. The system achieved a mean reprojection error of 0.210 pixels, indicating a precise result between the mathematical model and the physical characteristics of the lens.

3) Stereo Calibration and Rectification : To obtain the stereo extrinsic parameters, we capture a series of chessboard image pairs using the Lucia stereo system. The chessboard is placed at varying orientations and spatial positions, ensuring that each stereo pair provides distinct geometric constraints. These image pairs are processed using OpenCV to estimate the relative rotation and translation matrices between the two

TABLE III
INTRINSIC PARAMETERS AND DISTORTION COEFFICIENTS OF THE LUCIA CAMERA. THE CALIBRATION FOLLOWS THE PINHOLE CAMERA MODEL WITH BROWN–CONRADY DISTORTION.

Category	Symbol	Value
Pinhole Intrinsics (pixels)		
Focal length (horizontal)	f_x	607.26
Focal length (vertical)	f_y	606.80
Principal point (x)	c_x	486.18
Principal point (y)	c_y	636.19
Radial Distortion Coefficients		
Radial distortion	k_1	-1.9118
	k_2	-3.5315
	k_3	8.7469
Tangential Distortion Coefficients		
Tangential distortion	p_1	-0.00085
	p_2	0.00171
Calibration Accuracy (pixels)		
Mean reprojection error (RMSE)	–	0.210

(a) Sample visual capture of the calibration chessboard (b) Detected inner corners of the chessboard pattern

Fig. 4. Monocular camera calibration using a planar chessboard pattern.

Lucia pins.

The calibration quality was evaluated using two primary metrics: the global stereo reprojection error and the vertical epipolar error after rectification.

Stereo calibration results are summarized in Table IV. We achieved a mean reprojection error (E_{rms}) of **0.247** pixels across the valid stereo pairs, indicating accurate estimation of both intrinsic and extrinsic parameters. Furthermore, the quality of stereo rectification was assessed by measuring the residual vertical disparity between corresponding features. The mean epipolar alignment error (E_v) was calculated to be **0.188** pixels, confirming that the rectified image pairs are consistently row-aligned and geometrically coherent.

To further contextualize the geometric consistency of LuciaPin, we conducted the same calibration evaluation procedure on an industrial stereo module, the **Intel RealSense Depth Camera D405**. Under identical processing conditions, the D405 achieved a stereo RMS reprojection error of 0.136 pixels and a vertical epipolar alignment error of 0.037 pixels. While the factory-calibrated module exhibits lower residuals, LuciaPin remains well within the sub-pixel regime, demonstrating stable and reliable stereo geometry.

TABLE IV
QUANTITATIVE EVALUATION OF THE STEREO SYSTEM CALIBRATION. THE LOW RMS REPROJECTION ERROR AND VERTICAL EPIPOLAR ALIGNMENT ERROR INDICATE HIGH GEOMETRIC CONSISTENCY.

Metric	Symbol	Value	Unit
Image resolution	$W \times H$	960×1280	pixels
Focal length	f	632.23	pixels
Baseline distance	B	70.06	mm
Mean reprojection error	E_{rms}	**0.247**	pixels
Epipolar alignment error	E_v	**0.188**	pixels

Importantly, the objective of LuciaPin is not to surpass integrated industrial depth cameras in raw geometric residuals, but to provide a configurable dual-visualsensor platform within an open SDK framework. The measured baseline of LuciaPin (70 mm) is substantially larger than that of the D405 (18 mm), offering increased disparity sensitivity in mid-range stereo reconstruction while maintaining geometric stability. These results indicate that the calibration accuracy of LuciaPin is sufficient to support downstream AI-driven perception and reconstruction tasks within the proposed integration framework.

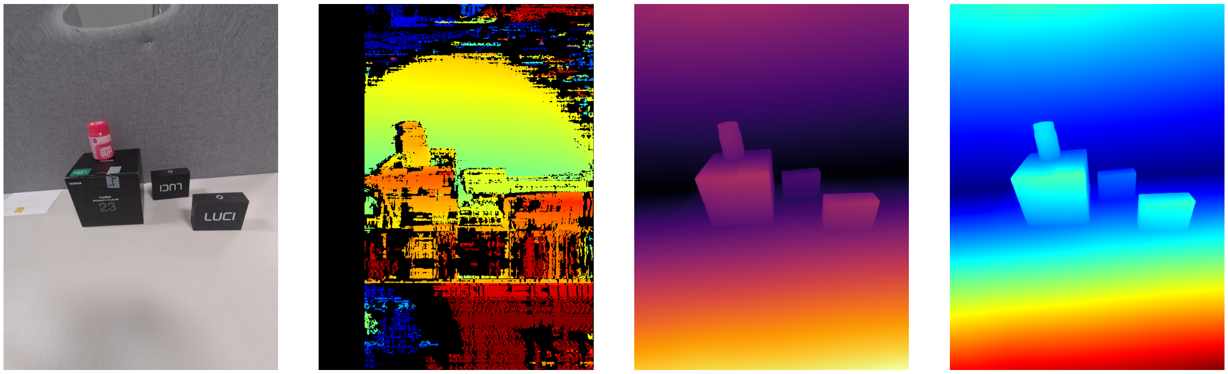
4) Stereo Depth Calculation and Fusion: To retrieve 3D geometry and depth information from the captured stereo pairs of images, we set up a hybrid approach that integrates traditional geometric calculation method with learning-based depth-estimation algorithm.

First, we apply the Semi-Global Block Matching [41](SGBM) algorithm to the rectified image pairs. Using the calibrated baseline B and focal length f, SGBM method illustrates how to compute the disparity d to recover the absolute metric depth $Z = f \cdot B/d$. Although geometrically accurate, the method is sensitive to textureless regions,such as white walls or reflective surfaces—where matching ambiguities frequently cause significant noise or data bias.

To avoid these limitations, we incorporate **Depth Anything V2** [42], a SOTA model for monocular depth estimation. This model performs well in extracting dense, edge-preserving depth structures even in challenging lighting and low-texture scenarios. However, its output represents relative depth without physical scales.

Therefore, we implement a fusion pipeline to combine the complementary advantages of both methods. The depth estimation results are visualized in Fig. **??**. As shown in Fig. 5a, it displays the rectified left view. The process removes distortion to restore natural geometry, providing a consistent baseline for the following steps. While raw stereo matching (Fig. 5b) captures the correct physical scale, it leaves significant voids in textureless regions such as the cabinet surfaces. In contrast, the monocular estimate from Depth Anything V2 (Fig. 5c) yields a structurally complete map with sharp edges, while it lacks absolute metric information.

To synthesize the final result, we align the prediction of left luci's view to the sparse stereo anchors using a robust RANSAC [43] linear fit. The resulting fused depth map (Fig. 5d) effectively recovers dense 3D geometry by combining the fine details from the AI model with the real physical



(a) Reference RGB (b) Sparse Stereo (SGBM) (c) DepthAnythingV2 (d) Dense Metric Fusion

Fig. 5. Comparison of RGB input, sparse stereo depth, monocular depth prediction, and the proposed dense metric fusion result.

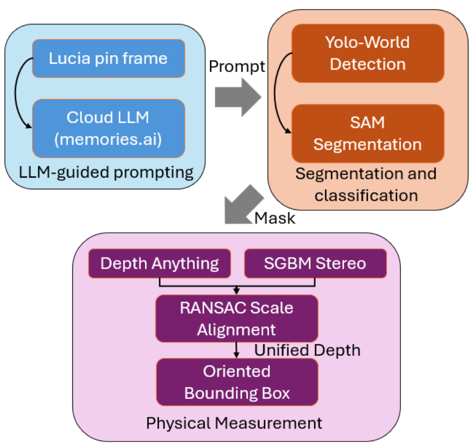
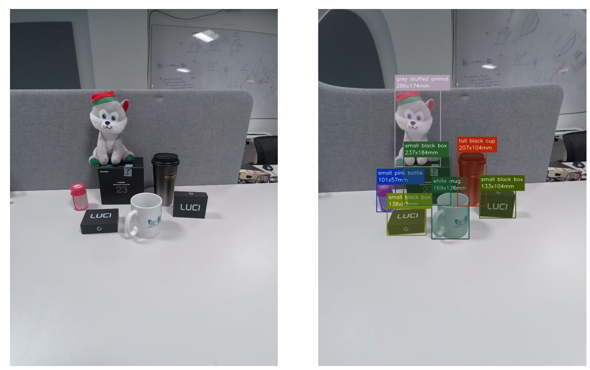


Fig. 6. Overview of the proposed pipeline. The Lucia Pin provides an LLM prompt generated by Memories.ai, which guides open-vocabulary object detection and segmentation. The resulting mask is then combined with stereo and monocular depth to estimate the object's physical size.



(a) Luci Pin's view of testing objects (b) Stereo visual result of objects

Fig. 7. Object measurement result using the proposed stereo pipeline.

scale provided by the stereo rig.

5) Stereo Object Classification and Size Estimation: Beyond serving as a monocular recorder, the Lucia Pin operates as an intelligent user agent. The core of our stereo Lucia system integrates real-time video perception with LLM-based interaction. To support this capability, we design a three-stage pipeline for object understanding, classification, and metric size estimation: (i) LLM-guided prompting, (ii) segmentation-based classification, and (iii) depth-driven physical measurement.

As illustrated in Fig. 6, frames captured by the Lucia Pin are transmitted to a cloud server hosted by Memories.ai, where an LLM generates a textual interpretation of the scene. Under normal operation, this feedback is spoken aloud by the device. In our pipeline, however, the returned description is instead used as the prompt for the subsequent perception module.

For object classification, we leverage YOLO-World [44], which supports detection conditioned directly on LLM-generated queries (e.g., "white mug," "black box"). This design allows the system to recognize arbitrary objects without the need for re-training. The coarse bounding boxes produced by YOLO-World are subsequently used as *spatial prompts* for the Segment Anything Model (SAM) [42], enabling the generation of clean, pixel-level masks. This coarse-to-fine procedure ensures accurate object boundaries, which are essential for reliable depth and shape extraction.

To estimate physical dimensions, as previous subsection illustrates, we compute a sparse metric depth map using SGBM, while a dense relative depth map is obtained using Depth Anything V2 [42]. Absolute scale is recovered by fitting a linear model using RANSAC:

$$d_{\text{metric}} = s \cdot d_{\text{mono}} + t \tag{1}$$

where s and t denote scale and offset parameters. This alignment allows missing or unreliable SGBM regions to be filled with metric-consistent depth derived from the monocular prediction. Because depth estimation is unreliable at object borders, we erode the segmentation mask with a 5×5 kernel to remove noisy boundary pixels before extracting depth values.

Finally, we determine the object's orientation and dimensions using Principal Component Analysis (PCA). By computing the covariance matrix of the object's 3D point cloud, we extract eigenvectors that represent the principal axes

TABLE V
OBJECT 3D DIMENSIONS (MM)

Object LLM prompts	$Height$	$Width$	$Thickness$
White mug	169.1	126.4	47.5
Pink bottle	101.2	56.6	43.3
Tall black cup	206.7	104.2	32.0
Stuffed animal	285.8	174.5	40.9
Black box(LUCI right)	133.1	104.0	17.7
Black box(LUCI left)	138.1	93.4	14.5
Black box (FUJI lense)	236.6	184.5	33.9

(Height(Y-direction of camera coordinate), Width(X-direction of camera coordinate), and Thickness(Z-direction of camera coordinate)). Projecting the points onto these axes yields a tight oriented bounding box (OBB), enabling accurate, pose-invariant measurement of the object's physical size.

IV. CONCLUSION

In Case B, we show that a dual-Lucia stereo setup can provide reliable near-range depth estimation while being augmented by LLM-driven scene understanding. The combination of language-guided perception and metric depth reconstruction enables object-level 3D reasoning and measurement, extending Lucia beyond passive image capture toward interactive spatial understanding.

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V. Biography Section

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